

Environmental Toxicology

Instructor: Lawrence K. Duffy, 246 WRRB, lkduffy@alaska.edu, 474-7525

Office Hours: Tuesday and Thursday following lectures. However students are encouraged to contact the instructor by phone or email at any time

Lecture: Tuesdays and Thursdays, 11:30 am – 1:00 pm. *Room 201B* Murie 103

Textbook: Fundamentals of Ecotoxicology 4th Edition (ISBN 978-14-666-58-2293) by Michael C Newman, CRC Press, Taylor and Francis Group. No supplementary reading required for purchase. Primary literature will be provided by instructor.

Course Description: This 3 credit course will discuss the basic components of environmental and ecotoxicology, and explores exposure of toxic chemicals to animals and plants and their impact on health and the ecosystem, using a One Health model. Environmental Toxicology will focus on the general properties and principles of persistent and/or toxic chemicals commonly encountered in air, water, fish and wildlife. Numerous natural and synthetic chemicals in the environment will be discussed from a global perspective with some bias towards arctic and subarctic regions. This toxicology course combines aspects of environmental science, vertebrate biochemistry and physiology and environmental chemistry in a manner to understand how systems are impacted and function.

Student Learning Outcomes:

- increase understanding of biochemistry with respect to environmental contaminants
- increase understanding of the chemical-biotic interactions
- gain an understanding of how biota alters the structure and dynamics of contaminants in the diverse ecosystems of the North

Course Goals:

This course will provide the basic foundations for Environmental Toxicology
 The class will focus on specific aspects of Environmental Toxicology that interest the student
 Develop an appreciation of the complex system of contaminant interactions in high latitude systems

Instructional Methods:

The teaching methods employed in this course will consist of lectures and "chalk talk" by the instructor. It is absolutely crucial that reading (see schedule) of sections is done in advance. Student essays and presentations on specific topics will also be used.

Grading:

Exams: There will be four term exams (50 points each); one for each major section of the course.
 Oral class participation in the form of discussions will be included in the final grade (100 points). This entails an active involvement into the regular lecture materials discussed. Quiz/Class participation (100 points).

Undergraduate Students

Exams	200 points (4 exams 50 points each)
Discussions / Research Proj.	200 points (Discussions: 50 points per discussion package) <i>100 points for Project</i>
Graduate total	600 points